

Data Collection and Preprocessing Phase

Date	26 April 2025
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Project Title	Global Malnutrition Trends: A Power BI Analysis (1983-2019)
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Section	Description
Data Overview	The dataset "Malnutrition Across the Globe" from Kaggle provides malnutrition statistics (stunting, wasting, underweight) for various countries and years (1983-2019). We will use it to analyze global malnutrition trends using Power BI.
Data Cleaning	Handle missing values by imputing using forward or backward fill from adjacent years. Identify and remove or correct outliers based on cross-validation with WHO/UNICEF data. Standardize inconsistent country names using a mapping table with ISO codes.
Data Transformation	Use Power Query Editor to filter data by year and region. Pivot data to create summary tables and unpivot it back for charting. Create calculated columns for new metrics (e.g., malnutrition prevalence change over time). Sort to highlight important data.
Data Type Conversion	Ensure all columns have the correct data types (e.g., numeric columns are set as decimal numbers, date columns are set as dates). Correct data types by using Power Query Editor for data validation and Power BI's transformation tools for accurate data handling
Column Splitting and Merging	No column splitting or merging is anticipated at this stage. If needed, create calculated columns for merged data (e.g., combining regions). Use split to split data into multiple columns.

Data Modeling	Define relationships between the country table, year table, and malnutrition data table. Create measures using DAX (e.g., average malnutrition rate, year-over-year change) to enable advanced analysis and create interactive visualizations.
Save Processed Data	Save the cleaned and processed data as a Power BI Desktop file (.pbix). Consider exporting cleaned data as a CSV file or Power BI dataflow for future use in other projects.